

Memory Gallops

Here's how seven brands of memory hit the turf to race against each other

For most computer buyers, all that concerns the RAM is how much of it there is. Most people do not know the brand, or the role it plays in the system. But once applications start loading slowly, games start freezing and the machine generally begins like a dimwit, the first thing that comes to mind is an upgrade. And the friendly neighbourhood computerwallah blames it all on the RAM. Thanks to these memory modules, you can load applications faster, play games smoothly and access data quickly. To shed more light on the mystical nature of the RAM, we concluded the first-of-its-kind RAM shootout.

Analysis

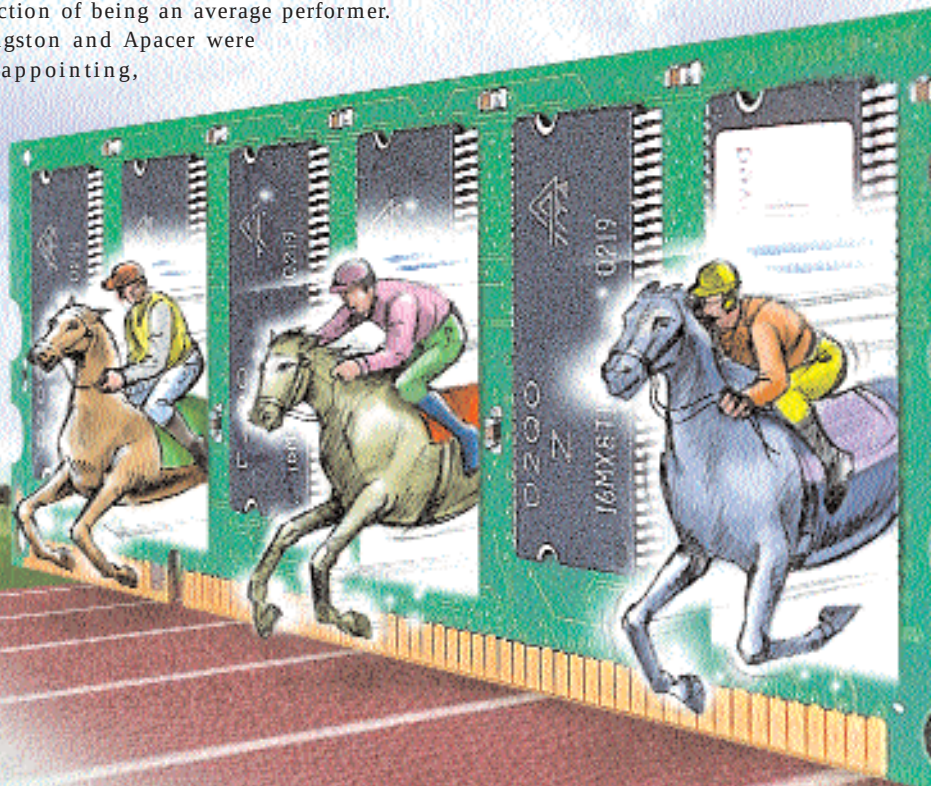
We had two categories—the horizontal and the vertical. In the horizontal category, we tested the different brands of RAM available, at the same data rate (333 MHz), while the CAS latency (*See box below: 'CAS Latency'*) was kept at a constant 2.5 ns. The vertical tests were dedicated to modules with Hynix chips, which we tested for all the data rates that they're currently available in. This gave an overview of the performance gain, with increase in data rate, in the same category of chip.

We received two high performance, DDR 400 TwinX XMS Low Latency memory modules from Corsair. We clocked one of them, at 333 MHz with the CAS latency kept at 2.5ns. We ran our tests on it, and the results were used as reference for the other modules. We used the underclocked Corsair as the reference because it indicates the maximum potential throughput of the DDR333 standard.

In the horizontal tests, in spite of the close competition, the Transcend module gave the best performance. Transcend uses PSC memory chips on its modules, which showed its mettle in the PC Mark, Adobe Photoshop and Content Creation tests. The Kingston and Apacer were the next best in terms of performance. We were surprised by the Hynix module, which scored well—Hynix has the dubious distinction of being an average performer. Kingston and Apacer were disappointing,

because of their higher-than-Hynix prices. The highest priced module was the Apacer, which uses Infineon memory chips. It stood third in the performance tests.

Pricing certainly affects these brands, since DDR 333 is catching up in the market. Unless prices are brought down, manufacturers like Kingston and Apacer can't hope to make a dent in a RAM market that's dominated by Hynix.



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CAS Latency

The RAM module is an organised collection of integrated chips (ICs). Each module is controlled by the memory controller, which handles the signals going from the CPU to the RAM. There are two signals—the Row Access Strobe (RAS) and the Column Access Strobe (CAS). Each memory chip is divided into rows and columns, making it look like a cell matrix. Each cell has a row address and

a column address. When the CPU sends a signal to the memory controller, it first accesses the row by putting an address on the memory's address pins, and activating the RAS signal. Then it waits a few clock cycles—this is the RAS-to-CAS Delay. It then puts the column address on the address pins, and activates the CAS signal. Then there's a wait for another few clock cycles, which is the CAS delay. Finally, the data

appears on the pins of the RAM.

The CAS delay is called CAS Latency. Lower CAS latency provides faster data access. CAS-2 is where the CPU waits for 2 clock cycles; CAS-3 is where it waits for 3 clock cycles, and so on. However, CAS-2 affords a performance gain over CAS-3 only when you play games or overclock your CPU. The CAS latency is usually 2.5 for DDR RAM.